



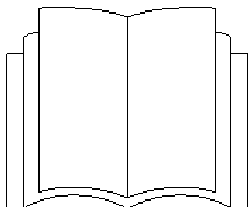
Annex for the original attachment's Instruction Manual



Log Cutter/Splitter

JAPA 300

Product number:	300-700-series loaders:	A21029
	200-series loaders:	A21262



Please read carefully the original attachment's instruction manual before using the equipment, as well as this annex, and follow all instructions.

This annex for the operator's manual includes specific information about the use of the equipment with Avant multi purpose loaders.

Keep the original operator's manual as well as this annex for later reference.

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This manual uses the following warning symbols to indicate important points that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:
This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This annex for the instruction manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for AVANT multi purpose loaders. The log cutter/splitter has been adapted for use with AVANT loaders by the original manufacturer of the equipment.

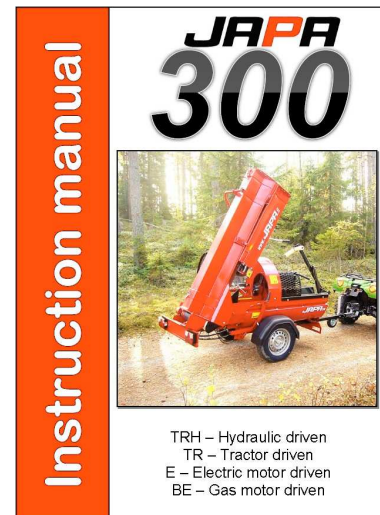
The original manual, provided with the log splitter/cutter, includes the information of safe use of the equipment. This annex for the manual includes important information of mounting and using the equipment with AVANT loaders.

By familiarising yourself with the original operator's manual and the additional instructions provided in this annex, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Ensure that all persons using the loader and the log cutter/splitter have received proper guidance and familiarised themselves with the manuals of the loader and attachment and all safety instructions before using the equipment.

Keep this annex of the manual at hand throughout the service life of the equipment. If this annex for the manual or the original operator's manual is lost or damaged, you can request a new one from your nearest AVANT dealer. If you sell or transfer the equipment, be sure to hand both parts of the manual to the new owner.



This annex for the original instruction manual of the log cutter/splitter machine does not cover all of the instructions needed for safe operation of the equipment. It is not meant to be distributed separately. The original instructions manual, provided with the machine, should be considered as the main source of information about the safety, use and the maintenance of the equipment. This annex includes important information about the use of the equipment specifically with AVANT multi purpose loaders.



2. Designed purpose of use

The JAPA firewood machines are hydraulically operated attachments designed and manufactured to be suitable for use with 200-700 –series AVANT multi purpose loaders. They are intended to be used by one person at a time for cutting firewood and splitting it into halves or quarters. The log cutter/splitter cuts the log with a chain saw blade to the adjusted length, and then the log is split with a hydraulic cylinder, which presses the log against the splitting wedge.

The log cutter/splitter is meant to be attached on the AVANT loader during use. The attachment is powered by the hydraulic outlet of the Avant loader. The log conveyor, shown in the original manual, is an optional extra. Contact your AVANT dealer for further information about its availability.

Familiarise yourself with the instruction manual's instructions regarding service and maintenance. Demanding repair and maintenance operations, such as hydraulic repairs, should be left for professional maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the log splitter with Avant loaders

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



- When mounting the log cutter/splitter, ensure that the **locking pins** of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
- The log cutter/splitter is to be used by one operator at a time. Keep others at a safe distance, and do not let others to operate the controls of the loader while using the attachment.
- When transporting the equipment, pay attention to the contours of the terrain as well as other hazards, such as loose rocks or branches and trees reaching the driver's area.
- Always transport the log cutter/splitter as low and close to the machine as possible to keep the centre of gravity low. Keep the telescopic boom retracted.
- Keep hands and feet away from moving parts.
- When detaching the log cutter/splitter, ensure that it is secured against accidental movement.
- Never perform any maintenance or adjustments while the loader engine is running.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual. Get to know the loader controls and how to drive it safely.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under or in front of raised boom or attachment.



Note the effects of heavy loads to the the stability and handling characteristics and of the loader, especially when operating with the smaller loader models. Always keep the load close to the ground and keep the telescopic boom retracted. Drive slowly and carefully and avoid steep slopes. Use additional counterweights as needed. Read the operator's manual of the loader.



Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection.
- The use of safety boots is recommended while working with the loader.



- Wear protective gloves.

4. Technical specifications

On the original instruction manual are listed the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer.

Table 1 –Log cutter/splitter specifications

Product number	A21262	A21029
Log length, min-max:	200 - 500 mm	
Log diameter, min-max:	20 - 300 mm	
Weight:	265 kg	
Minimum operating temperature:	-25°C	
Saw flange:	13"	
Splitting force:	30 kN	
Dimensions, transport configuration: L x W x H:	2780 x 1030 x 2450 mm	
Maximum input of hydraulic energy:	40 l/min, 22,5 MPa (225 bar)	
Suitable AVANT loader:	216, 218, 220	320-750

5. Assembly of the log cutter/splitter

The log cutter/splitter is ready for use as delivered. Attach the attachment and its hydraulic hoses according to the following instructions.

5.1. Attaching the log cutter/splitter to the loader

Mounting the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may fall on the operator or slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions shown in the following pages. Also remember the safety instructions in Section 3 and the information in the original instruction manual.

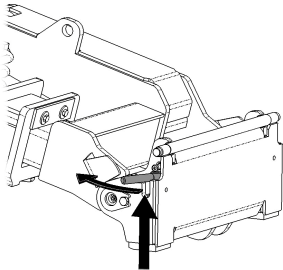
The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic, but due to the relatively high position of the quick attach plate on the log splitter, and the risk of tipping over of the attachment, the locking pins must be locked manually.



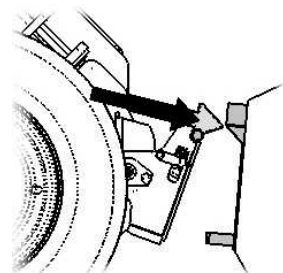
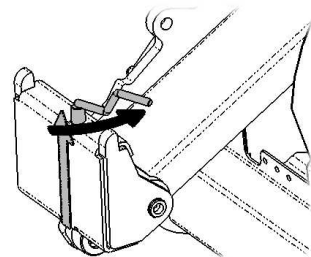
Never move or raise an attachment that has not been locked.

The attachment is mounted to the loader as follows:

Step 1:

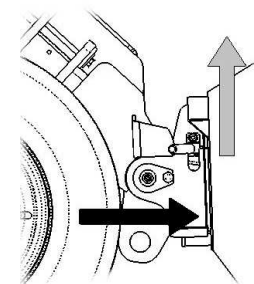
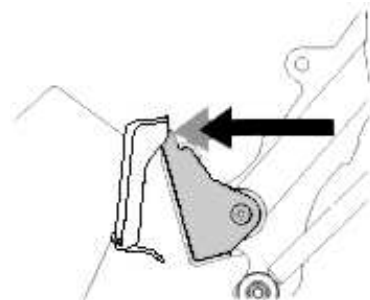


- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your 400-700 series loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.



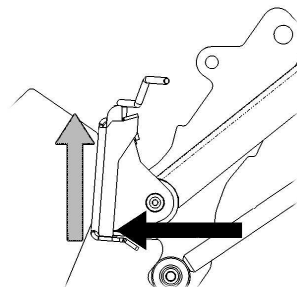
Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

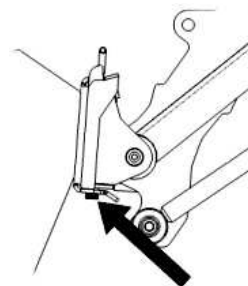
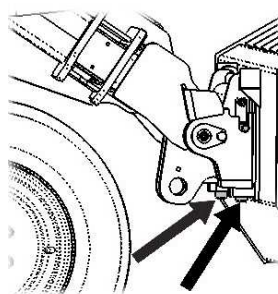


Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually as you connect the hydraulic fittings.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may tip on the boom or operator, and cause injuries.



Do not use the automatic return of the locking pins with the log cutter/splitter. Because the quick attach plate of the machine is at high level, tilting of an unlocked log cutter/splitter increases the risk of toppling over. We recommend that you lower the log splitter onto the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic couplings

The attachment is equipped with either standard quick couplings or the multiconnector system. If you want to change the type of the hydraulic couplings for the 300-700 series loaders, contact your Avant dealer or service point for instructions or installation services. The log cutter/splitter A21262 for 200-series loaders is always equipped with standard hydraulic quick couplings.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

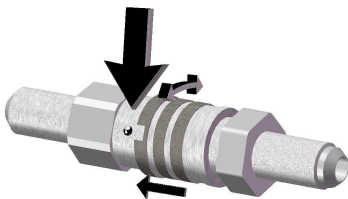
Connecting standard quick couplings:

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times and also move the control levers of the attachment. This way, there will be no residual pressure in the system and it is easier to connect the fittings.

To connect and disconnect the fittings, move the collar at the end of the female fitting. The hoses should be connected so that the male fitting of the attachment is connected on the loader to the fitting with coloured cap. The second hose is connected on the single male fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt.



In a 200-series machine, the hoses can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.

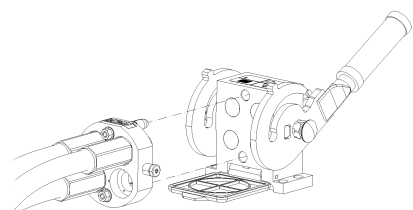


In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down.

Connect and lock the multiconnector by turning the lever towards the loader. The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control levers of the attachment to neutral position and also turn the auxiliary hydraulics control lever of the loader to its neutral position.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the control levers of the attachment and the auxiliary hydraulics control lever of the loader, when the engine is turned off.

6. Operation

Once the log cutter/splitter has been mounted according to the instructions and the hydraulic hoses are connected, it is ready for work. Read the original operator's manual of the equipment and also read the following instructions about work site, adjustments and use of the log splitter.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.

6.1. Setting up the log cutter/splitter

- Transport the log cutter/splitter carefully and note the effect of weight at the front to the stability of the loader.
- Choose a level working site, which can carry the load of the equipment and the logs.
- The operating area must be even and it must not be slippery. Clear the area from any obstacles that could cause a tripping hazard.
- Operate the log cutter/splitter in daylight or good artificial light and in well ventilated areas.
- Check regularly to ensure there are no hydraulic leaks. Refer to original instruction manual for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Make sure to lower the attachment securely on the ground. Ensure that the parking brake of the loader is engaged and that the splitter can not move during use.



6.2. Use



When operating the log splitter, the loader must be equipped with a hydraulic oil cooler. Without the cooler hydraulic oil may overheat. Overheating of the oil decreases the power and affects the service life of the hydraulic components of both the attachment and the loader, and may also lead to more serious damage to the hydraulic systems. Regularly clean the loader oil cooler core; refer to the loader manual for more instructions.

To start the log cutter/splitter, turn the auxiliary hydraulics lever on the loader to its locked position and set moderate engine rpm, depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Do not exceed the maximum permissible input of hydraulic oil, full rpm must not be used when operating with 600 or 700 series loaders. Note that unnecessarily high engine rotating speed will not improve the splitting force. Operate the log cutter/splitter and perform maintenance procedures as instructed in the original manual of the equipment.



Check the operation and the adjustments of the splitter with a few empty cycles before loading a log.

The lubrication of the log cutter chain saw consumes a small amount of hydraulic oil from the loader. Check the hydraulic oil level regularly.



The log splitter is intended to be used by one operator only.



Never leave the machine unattended with the engine running.



Do not use the attachment for any other purpose than for cutting and splitting firewood.



Read the original instruction manual for any further information of adjustments, use and maintenance of the equipment.

7. Maintenance and warranty

Instructions for maintenance and service are provided with the original instruction manual as well as the spare parts list. Contact your AVANT dealer for parts and service.

The manufacturer of the log cutter/splitter grants the warranty of one year for the log cutter/splitter. Warranty terms are provided with the instruction manual. For warranty claims, contact your AVANT dealer for further information.

The manufacturer of the log cutter/splitter provides the declaration of conformity for the equipment.



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